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Continuous torque transmission system

Abstract

A Continuous Torque Transmission (CTT) system that includes a first planetary gear assembly, a first electric motor-generator, a control module, a second electric motor-generator, and a third electric motor-generator. The first electric motor-generator includes a first stator and a first rotor. The second electric motor-generator includes a second stator and a second rotor, and the third electric motor-generator includes a third stator and a third rotor. A rectifier is electrically coupled to a control module and is configured to receive alternating current (AC) and convert it into direct current (DC). The control module may configure the CTT system in an acceleration mode or a deceleration mode, via a mode switch. In the acceleration mode, the first stator acts as a generator. In the deceleration mode, the third stator acts as a generator.

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Background/Summary

BACKGROUND

Field of the Invention

(1) The present invention relates, generally, to hybrid transmission systems, and more particularly to a Continuous Torque Transmission (CTT) system for a vehicle for selectively combining mechanical and electrical power.

Description of the Prior Art

(2) The development of systems to combine mechanical and electrical power dates back to the early 20th century with the advent of hybrid technologies. Initially, these systems were rudimentary, involving basic methods to switch between mechanical and electrical drives. Over time, advancements in electronics, materials science, and control systems have led to more sophisticated and efficient solutions. Traditional hybrid power transmission systems often suffer from inefficiencies when switching between or combining mechanical and electrical power sources. These inefficiencies can lead to energy losses, reduced performance, and increased wear and tear on system components. The challenge lies in developing a system that can continuously and smoothly combine these power sources to deliver optimal torque and power without interruption or significant energy loss.

(3) One of the conventional hybrid systems for power transmission in vehicles is the ‘parallel hybrid system’ in which both an internal combustion engine (mechanical) and an electric motor (electrical) drive the vehicle's wheels. However, switching between these power sources can cause interruptions and inefficiencies. Another conventional hybrid system called ‘series hybrid system’ uses an electric motor to drive the wheels, with an internal combustion engine generating electricity to power the motor. While this reduces the mechanical complexity, it can still result in energy conversion losses.

(4) There is, therefore, a need for solutions that enable the smooth and continuous blending of mechanical and electrical power sources, ensuring no interruptions in power delivery, maximize the efficiency of power transmission by reducing energy losses associated with switching and combining power sources, and increase the overall performance of the system by optimizing the torque delivery and maintaining consistent power output.

SUMMARY

(5) In one aspect, a Continuous Torque Transmission (CTT) system is disclosed. The CTT system may include a first planetary gear assembly that may include a first ring gear attached to an input shaft rotating at a consistent speed, a first sun gear attached to a first rotor, and a set of first planet gears, each engaging with the first ring gear and the first sun gear. The CTT system may further include a first electric motor-generator that may include a first stator attached to the set of first planet gears and a first rotor electromagnetically coupled with the first stator. The CTT system may further include a control module electrically connected to the first stator and the set of first planet gears. Further, the CTT system may include a second electric motor-generator that may further include a second stator connected to the set of first planet gears, the first stator, and the control module and a second rotor electromagnetically coupled with the second stator. The CTT system may further include a third electric motor-generator that may further include a third stator and a third rotor. The third stator may be connected to the set of first planet gears, the first stator, the control module, and the second stator. The set of first planet gears, the first stator, the control module, the second stator, and the third stator may be coupled to an output shaft. The CTT system may further include a third rotor configured to electromagnetically coupled with the third stator, and a rectifier electrically coupled to control module and the configured to receive alternating current (AC) and convert into direct current (DC). The control module may configure the CTT system in one of an acceleration mode and a deceleration mode, via the mode switch. In the acceleration mode, the first stator may act as a generator to generate electric power that is supplied to the rectifier. In the deceleration mode, the third stator may act as a generator to supply electric power to the rectifier. The rectifier may supply electric power to the first stator that acts as a motor to maintain a constant speed of the input shaft during the deceleration of the output shaft.

(6) In an embodiment, the second rotor may be fixed to a housing associated with the CTT system to fix rotation of the second rotor.

(7) In an embodiment, the CTT system may further include a second planetary gear assembly that may include a second ring gear attached to the set of first planet gears, the first stator, the control module, the second stator, and the third stator, and coupled to the output shaft. Further, the CTT system may include a second sun gear attached to the third rotor and a set of second planet gears, each engaging with the second ring gear and the first sun gear.

(8) In an embodiment, the set of second planet gears may be attached to a shaft, such that the shaft may be adapted to rotate in one direction.

(9) In an embodiment, the CTT system may further include a first spur gear attached to the second ring gear, such that the first spur gear may be further coupled to an output shaft. Further, the first spur gear may be coupled to the output shaft. A second spur gear and a third spur gear may be attached to a common connecting shaft. A fourth spur gear may be attached to the output shaft, such that the fourth spur gear is offset from the second and the third spur gear and engaging with the third spur gear.

(10) In an embodiment, when the output shaft is stationary: the rotation of the input shaft may cause the rotation of the first sun gear via the first ring gear and the set of first planet gears, to induce maximum voltage in the first stator; revolution of the set of first planet gears may be constrained; and the control module ceases the supply of electrical power to the second stator or the third stator.

(11) In an embodiment, when the output shaft is in the acceleration mode from being stationary: for a high torque application, the control module may direct electrical power to the third stator; or for a low torque application, the control module may direct electrical power to the second stator.

(12) In an embodiment, when the output shaft rotates at a medium speed range: the torque generated at the output shaft may be partly from the input shaft and partly from one of the second stator and the third stator.

(13) In an embodiment, when the output shaft rotates at a high-speed range: the first sun gear may

rotate at a lower speed; the first rotor rotating at the lower speed may induce a low voltage in the first stator; and the control module may cause the supply of electrical power to one of the second stator and the third stator.

(14) In an embodiment, when the output shaft rotates at the high-speed range and is in the deceleration mode, the output shaft may rotate the third stator and the second stator, and the control module may cause the supply of electrical power to the first stator.

(15) In an embodiment, the control module may include a mode switch configured to switch the CTT system between the acceleration mode and the deceleration mode, based on an instruction.

(16) In an embodiment, the control module may further include a receiver configured to wirelessly receive the instruction and transmit the instruction to the mode switch.

(17) In an embodiment, the CTT system may further include a voltage regulator coupled to the control module. The voltage regulator may be configured to receive electricity supply from the rectifier and regulate the voltage of electricity supplied to the control module.

(18) In an embodiment, the control module may include a first motor controller configured to receive electrical power from the voltage regulator and the instruction from the receiver, to control the third stator in the acceleration mode, or control the first stator in the deceleration mode. The control module may further include a second motor controller configured to receive electrical power from the voltage regulator and the instruction from the receiver, to control the second stator.

(19) In an embodiment, in the acceleration mode, the first motor controller may be coupled with the third stator, and the second motor controller may be connected with the second stator. In the deceleration mode, the first stator may be coupled with the first motor controller.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates a schematic diagram of a Continuous Torque Transmission (CTT) system, according to an example embodiment of the disclosure.

(2) FIG. 2A illustrates another schematic diagram of the CTT system with a mode switch configured in a first configuration, according to an example embodiment of the disclosure.

(3) FIG. 2B illustrates a yet another schematic diagram of the CTT system with the mode switch configured in a second configuration, according to an example embodiment of the disclosure.

(4) FIG. 3 illustrates a flowchart of a method, according to an example embodiment of the disclosure.

DETAILED DESCRIPTION

(5) Referring to FIG. 1, a schematic diagram of a Continuous Torque Transmission (CTT) system **100** is illustrated, according to an example embodiment of the disclosure. The CTT system **100** may be implemented, for example, in a vehicle with an internal combustion engine, and provides for an efficient combining of electrical and mechanical power towards torque transmission.

(6) The CTT system **100** may include a first planetary gear assembly **104**, a first electric motor-generator **106**, a control module **108**, a second electric motor-generator **110**, a third electric motor-generator **112**, and a rectifier. The CTT system **100** may be coupled with an input shaft **102** via the first planetary gear assembly **104**. In particular, the first planetary gear assembly **104** may include a first ring gear **104A** that may be coupled to the input shaft **102**. The input shaft **102** may supply mechanical power and rotate at a consistent speed. The first planetary gear assembly **104** may further include a first sun gear **104B** and a set of first planet gears **104C**. As shown in FIG. 1, the first ring gear **104A** may engage with the set of first planet gears **104C**, and the set of first planet gears **104C** may further engage with the first sun gear **104B**. The set of first planet gears **104C** may include a plurality of gears planet gears, i.e., two or more planet gears.

(7) The first electric motor-generator **106** may include a first stator **106A** and a first rotor **106B**.

The first electric motor-generator **106** may be implemented as an electromechanical device that can function both as a motor, converting electrical energy into mechanical energy, and as a generator, converting mechanical energy into electrical energy. The first stator **106A** may be the stationary part of the first electric motor-generator **106**, and may generate a magnetic field which interacts with the first rotor **106B**. The first stator **106A** may include windings or permanent magnets mounted on a laminated iron core. Further, the first stator **106A** may have winding configurations corresponding to three-phase or single-phase configuration. The first rotor **106B** may be the rotating part of the first electric motor-generator **106** and may rotate due to the interaction with the magnetic field generated by the first stator **106A**. The first rotor **106B** may be made of coils of wire (wound rotor) or a solid metal (squirrel-cage rotor) encased in laminations to reduce energy loss, or permanent magnets in a specific order and number depending on stator **106A** configuration. For example, the first rotor **106B** may be a wound rotor or an induction (squirrel-cage) rotor or a permanent magnet assembly held together by a solid structure. As will be further appreciated by those skilled in the art, when operating as a motor, electrical energy is supplied to the windings of the first stator **106A**, creating a rotating magnetic field. This field induces a current in the rotor, causing it to spin and produce mechanical motion. When functioning as a generator, mechanical energy rotates the first rotor **106B** within the magnetic field, inducing an electric current in the windings of the first stator **106A**, thus generating electricity.

(8) The first rotor **106B** of the first electric motor-generator **106** may be attached to the first sun gear **104B** of the first planetary gear assembly **104**. Further, the first stator **106A** may be attached to the set of first planet gears **104C**. Furthermore, as mentioned above, the first rotor **106B** may be electromagnetically coupled with the first stator **106A**.

(9) The control module **108** may be electrically and physically connected to the first stator **106A** and the set of first planet gears **104C**. Further, the control module **108** may be free to rotate with the first planet gears **104C** and first stator **106A**.

(10) The second electric motor-generator **110** may include a second stator **110A** and a second rotor **110B**. Similar to the first electric motor-generator **106**, the second electric motor-generator **110** may also be implemented as an electromechanical device that can function both as a motor and as a generator. The second rotor **110B** may be the rotating part and may rotate due to the interaction with the magnetic field generated by the second stator **110A**. As such, when operating as a motor, electrical energy is supplied to the windings of the second stator **110A**, creating a rotating magnetic field that induces a current in the second rotor **110B**, causing it to spin and produce mechanical motion. When functioning as a generator, mechanical energy rotates the second rotor **110B** within the magnetic field, inducing an electric current in the windings of the second stator **110A**, thus generating electricity.

(11) In an embodiment, the second rotor **110B** may be fixed to a housing associated with the CTT system **100**. As such, the rotation of the second rotor **110B** may be fixed. Further, the second stator **110A** may be connected to each of the set of first planet gears **104C**, the first stator **106A**, and the control module **108**. In other words, the second stator **110A**, the set of first planet gears **104C**, the first stator **106A**, and the control module **108** may be attached to each other. Furthermore, as mentioned above, the second rotor **110B** may be electromagnetically coupled with the second stator **110A**.

(12) The third electric motor-generator **112** may include a third stator **112A** and a third rotor **112B**. Similar to the first electric motor-generator **106** and the second electric motor-generator **110**, the third electric motor-generator **112** may also be implemented as an electromechanical device that can function both as a motor and as a generator. The third rotor **112B** may be the rotating part and may rotate due to the interaction with the magnetic field generated by the third stator **112A**. As such, when operating as a motor, electrical energy is supplied to the windings of the third stator **112A**, creating a rotating magnetic field that induces a current in the third rotor **112B**, causing it to spin and produce mechanical motion. When functioning as a generator, mechanical energy rotates

the third rotor **112B** within the magnetic field, inducing an electric current in the windings of the third stator **112A**, thus generating electricity.

(13) In some embodiments, each of the first electric motor-generator **106**, the second electric motor-generator **110**, and the third electric motor-generator **112** may be a brushless motor, and therefore, may require three electrical wires to power them.

(14) The third stator **112A** may be connected to the set of first planet gears **104C**, the first stator **106A**, the control module **108**, and the second stator **110A**. As such, the third stator **112A**, the set of first planet gears **104C**, the first stator **106A**, the control module **108**, and the second stator **110A** may be connected to each other. Further, the set of first planet gears **104C**, the first stator **106A**, the control module **108**, the second stator **110A**, and the third stator **112A** may be coupled to an output shaft **118**. Furthermore, as mentioned above, the third rotor **112B** may be electromagnetically coupled with the third stator **112A**.

(15) The rectifier (not shown in FIG. 1; refer FIGS. 2A-2B) may be electrically coupled to the control module **108**. The rectifier may be configured to receive alternating current (AC) from an AC supply and convert it into direct current (DC). The rectifier may be a half-wave type rectifier or full-wave type rectifier. The half-wave type rectifier may convert only one half of the AC cycle to DC, and may use a single diode. The full-wave type rectifier may convert both halves of the AC cycle to DC, and have higher efficiency.

(16) The control module **108** may configure the CTT system **100** in an acceleration mode and or a deceleration mode. In the acceleration mode, the first stator **106A** may act as a generator to generate electric power. The generated electric power may be supplied to the rectifier (for example, a 3-phase bridge rectifier). In other words, when the output shaft is accelerating (for example, when the vehicle starts to move from being stationary), the first stator **106A** (i.e. the first electric motor-generator **106**) may act as generator to generate electric power. As such, the first rotor **106B** is rotated by the mechanical power from the input shaft **102**, to thereby configure the first electric motor-generator **106** as generator to generate electrical power. The rectifier may convert the alternating current (AC) into direct current (DC) that may be supplied to a voltage regulator (not shown in FIG. 1; refer FIGS. 2A-2B). The voltage regulator may bring the current to a constant voltage and send the current to a first motor controller and a second motor controller. Further, depending on a signal from a receiver, power may be sent to the second stator **110A** or the third stator **112A**. In particular, the power may be sent to the second stator **110A** for low torque demand, and to the third stator **112A** for high torque demand.

(17) During the deceleration mode, energy may be supplied to the CTT system **100** of the vehicle via the output shaft **118**. The energy may later be provided to the engine via the input shaft **102**. The mode switch (not shown in FIG. 1, refer FIGS. 2A-2B) may switch terminals. In the deceleration mode, the third stator **112A** may act as a generator to supply electric power to the rectifier. The rectifier may in turn supply the generated electric power to the first stator **106A**. In other words, when the output shaft is decelerating (for example, during braking of the vehicle), the third stator **112A** (i.e. the third electric motor-generator **112**) may act as generator to generate electric power. As such, the third rotor **112B** is rotated by the mechanical power from the output shaft **118**, to thereby configure the third electric motor-generator **112** as generator to generate electrical power. Further, in the deceleration mode, the first stator **106A** may act as a motor. Therefore, the first stator **106A** is supplied electrical power to cause rotation of the first rotor **106B**. As a result, a constant speed of the input shaft **104** is maintained during the deceleration of the output shaft **118**. The current generated from the third stator may enter the rectifier, then to the voltage regulator, and finally to the first motor controller. The first motor controller, with the help of the receiver may power the first stator **106A** so as to maintain a gear ratio that allows the input shaft **102** to run at a constant speed, even when the output shaft **118** is slowing down.

(18) In an embodiment, the CTT system **100** may further include a second planetary gear assembly **114**. In particular, the third stator **112A** of the third electric motor-generator **112** may be coupled

with the output shaft **118** via the second planetary gear assembly **114**. The second planetary gear assembly **114** may include a second ring gear **114A** that may be attached to the set of first planet gears **104C**, the first stator **106A**, the control module **108**, the second stator **110A**, and the third stator **112A**. Further, the second ring gear **114A** may be coupled to the output shaft **118**. The second planetary gear assembly **114** may further include a second sun gear **114B** that may be attached to the third rotor **112B**. Further, the second planetary gear assembly **114** may include a set of second planet gears **114C**, each of which may engage with the second ring gear **114A** and the first sun gear **104B**.

(19) In an embodiment, the set of second planet gears **114C** may be attached to a connecting shaft **120**, such that the connecting shaft **120** is configured to rotate in one direction. In other words, the connecting shaft **120** may be coupled to a one-direction mechanism that allows rotation of the connecting shaft **120** and hence the set of second planet gears **114C** only in direction and restricts the rotation in the opposite direction.

(20) The CTT system **100** may further include a gear assembly **116** that may couple the second ring gear **114A** to the output shaft **118**. As shown in FIG. 1, the gear assembly **116** may include a first spur gear **116A** that may be attached to the second ring gear **114A**. The first spur gear **116A** may be further coupled to an output shaft **118**. In particular, the gear assembly **116** may further include a second spur gear **116B**, a third spur gear **116C**, and a fourth spur gear **116D**. The second spur gear **116B** and the third spur gear **116C** may be attached to a common connecting shaft **116E**. The fourth spur gear **116D** may engage with the third spur gear **116C**. Further, the fourth spur gear **116D** may be attached to the output shaft **118**. As shown in FIG. 1, the fourth spur gear **116D** may be offset from the second spur gear **116B** and the third spur gear **116C**.

(21) Therefore, it should be noted that the set of first planet gears **104C** may be solidly mounted to the first stator **106A**, the control module **108**, the second stator **110A**, and the third stator **112A**, the second ring gear **114A** and the first spur gear **116A**. In particular, the input shaft **102** may be solidly mounted to the first ring gear **104A**. the first sun gear may be solidly mounted to the first rotor **106B**. The second sun gear **114B** may be solidly mounted to the third rotor **112B**. The second spur gear **116B** and the third spur gear **116C** may be solidly mounted on the same shaft **116E** and therefore spin at the same speed. The fourth spur gear **116D** may be solidly mounted to the output shaft **118**. The set of first planet gears **104C** may be connected to the output shaft **118** via the first spur gear **116A**, the second spur gear **116B**, the third spur gear **116C**, and the fourth spur gear **116D**. As such, the speed of the set of first planet gears **104C** may be proportional to the speed of the output shaft **118**. Further, the speed ratio between the first planet gears **104C** and the output shaft **118** may depend on the gear ratio of the first spur gear **116A** through the fourth spur gear **116D**.

When Output Shaft is Stationary

(22) During operation of the CTT system **100**, when the output shaft **118** is stationary: the rotation of the input shaft **102** may cause the rotation of the first sun gear **104B** via the first ring gear **104A** and the set of first planet gears **104C**, to induce maximum voltage in the first stator **106A**. Further, when the output shaft **118** is stationary, revolution of the set of first planet gears **104C** may be constrained. Furthermore, the control module **108** may cease the supply of electrical power to the second stator **110A** or the third stator **112A**.

(23) For example, when the vehicle and the output shaft **118** are stationary and the engine of the vehicle is running at 2000 RPM and hence the input shaft **102** is running at 2000 revolutions per minute (RPM), the first ring gear **104A** may also rotate at 2000 RPM. Since the speed of the output shaft **118** is 0, the speed of the set of first planet gears **104C** is also 0. As a result, in the current scenario, the first sun gear **104B** may be rotating at a maximum speed. For example, in a case scenario, a gear ratio between the first ring gear **104A** and the first sun gear **104B** is 4; therefore, the speed of the first sun gear **104B** and the first rotor **106B** will be 8000 RPM. This may induce maximum voltage in the first stator **106A**, for the control module **108** to operate the transmission.

The control module **108** may decrease the voltage to operate one of the second stator **110A** or the third stator **112A**. As soon as the vehicle starts driving, a signal may be transmitted (e.g. wirelessly) to the control module **108** which may then send electrical power to the third stator **112A** for high torque applications, or to the second stator **110A** for low torque applications.

When Output Shaft is Acceleration Mode from being Stationary (i.e. Vehicle Starts Driving)

(24) When the output shaft **118** is in the acceleration mode from being stationary (for example, when the vehicle started moving from being stationary), for a high torque application, the control module **108** may direct electrical power to the third stator **112A**. As will be understood, a high torque is required to move the vehicle from a stationary condition. Therefore, in such a scenario, the control module **108** may engage the third stator **112A** by directing electrical power to the third stator **112A**. It should be noted that while the third stator **112A** is suitable for high torque applications, the second stator **110A** may be more suitable for low torque applications. As such, for a low torque application (for example, when the vehicle is already moving), the control module **108** may direct electrical power to the second stator **110A**.

When Output Shaft is Rotating at Medium Speed Range

(25) In an embodiment, when the output shaft **118** is rotating at a medium speed range (for example, when the vehicle is moving at a medium speed), the torque generated at the output shaft **118** may be partly derived from the input shaft **102** and partly from one of the second stator **110A** and the third stator **112A**. In other words, in the medium speed range scenario, the output shaft **118** may be rotated by power delivered by a combination of the input shaft **102** (i.e. mechanical power from the engine) and power generated by one of the second stator **110A** and the third stator **112A** (i.e. using electricity).

(26) For example, when the engine speed and hence speed of the input shaft **102** is 2000 RPM, the first ring gear rotates at 2000 RPM. Further, since the set of first planet gears **104C** is rotating at medium RPM, speed of the first sun gear **104B** is also in medium range. In such a situation, half of the torque being supplied to the output shaft **118** is from the mechanical side (i.e. the vehicle engine via the input shaft **102**) and half from the electrical side (i.e. the second electric motor-generator **110** or the third electric motor-generator **112**).

When Output Shaft is Rotating at High-Speed Range (No Deceleration)

(27) When the output shaft **118** is rotating at a high-speed range and there is no deceleration, the first sun gear **104B** may rotate at a lower speed. Correspondingly, the first rotor **106B** may rotate at a lower speed. As such, the first rotor **106B**, rotating at lower speed, may induce a low voltage in the first stator **106A**. Further, the control module **108** may cause the supply of electrical power to one of the second stator **110A** and the third stator **112A**. Therefore, depending on low or high torque requirement, the control module **108** may supply electrical power to the second stator **110A** or the third stator **112A**, respectively. As such, one of the second stator **110A** and the third stator **112A** may generate power using electricity and supply to the output shaft **118** to facilitate rotation at the high speed range.

(28) For example, when the vehicle speed is high and the engine speed (and hence speed of input shaft **102**) is 2000 RPM, the speed of the set of first planet gears **104C** is close to the speed of the first ring gear **104A**. As a result, the first sun gear **104B** rotates in a low speed range. The first rotor **106B** rotates at low speed, inducing low voltage in the first stator **106A**. The control module **108** may increase the voltage to operate the second stator **110A** or the third stator **112A**. In such a scenario, most of the torque to the output shaft **118** is transmitted mechanically, i.e. from the input shaft **102**, as this is the most efficient way to transmit power.

When Output Shaft Rotates at the High-Speed Range (and is in Deceleration Mode)

(29) When the output shaft **118** is rotating at the high-speed range and is in the deceleration mode, the output shaft **118** may rotate the third stator **112A** and the second stator **110A**. As the output shaft **118** is decelerating, the power from the output shaft **118** may be used to generate electrical power via the third stator **112A** and the second stator **110A**. The generated electricity may be

supplied to the first stator **106A**, by the control module **108**. The rotation of the first stator **106A** may therefore be transferred to the input shaft **102** that may, for example, cause to charge a battery. (30) In the above scenario (i.e. when the vehicle is running at high speed range and is decelerating), for example, when the engine speed (and hence speed of input shaft **102**) is 2000 RPM, the vehicle can be decelerated using the engine of the vehicle. The power coming in through the output shaft **118** may drive the second electric motor-generator **110** and the third electric motor-generator **112**. The power obtained by the control module **108** may be converted and used to drive the first electric motor-generator **106**, to maintain a gear ratio that allow the engine to run at constant RPM while the vehicle decelerates.

(31) As mentioned above, the CTT system **100** may include the rectifier and the voltage regulator. In addition, the CTT system **100** may include a mode switch, a receiver, a first motor controller, and a second motor controller. These components are explained in detail, in conjunction with FIGS. 2A-2B.

(32) Referring now to FIGS. 2A-2B, schematic diagrams of the CTT system **100** are illustrated, in accordance with some embodiments. As shown, the CTT system **100** may include a rectifier **202**, a voltage regulator **204**, a first motor controller **206A**, a second motor controller **206B**, a mode switch **208**, and a receiver **210**. FIG. 2A illustrates a schematic diagram of the CTT system **100** with the mode switch **208** of the control module **108** configured in a first configuration; and FIG. 2B illustrates a schematic diagram of the CTT system **100** with the mode switch **208** configured in a second configuration. The control module **108** of the CTT system **100** may include the mode switch **208** that may be configured to switch the CTT system **100** between the acceleration mode and the deceleration mode, based on an instruction. For example, when configured in the first configuration, the mode switch **208** may configure the CTT system **100** in the acceleration mode. Further, when configured in the second configuration, the mode switch **208** may configure the CTT system **100** in the deceleration mode. The CTT system **100** may further include a rectifier **202**, and a voltage regulator **204**, first motor controller **206A**, and a second motor controller **206B**. For example, the rectifier **202** may be a 3-phase bridge rectifier that may receive alternating current (AC) through I1, I2, and I3 terminals and transform it into direct current (DC).

(33) The voltage regulator **204**, for example, may be coupled to the control module **108** and may be configured to receive electricity supply from the rectifier **202** and regulate the voltage of electricity supplied to the control module **108**. The first motor controller **206A**, for example, may be configured to receive electrical power from the voltage regulator **204** and the instruction from the receiver **210**, to control the third stator **112A** in the acceleration mode, or control the first stator **106A** in the deceleration mode. The second motor controller **206B**, for example, may be configured to receive electrical power from the voltage regulator **204** and the instruction from the receiver **210**, to control the second stator **110A**.

(34) As shown in FIGS. 2A-2B, the voltage regulator **204** may receive direct current (DC) from the rectifier **202** through terminals V1 and V2. The voltage of this direct current may be variable. For the first motor controller **206A** and the second motor controller **206B** to work, a constant voltage (for example, 32 Volts(V)) may be required. If the input voltage is lower than 32 V, the voltage regulator **204** may work as a boost converter, and may increase the voltage to 32 V. If the input voltage is higher than 32 V, the voltage regulator **204** may work as a buck converter, and decrease the voltage to 32 V. Thereafter, the voltage regulator **204** may supply the regulated voltage (e.g. 32 V) to the second motor controller **206B** through V3 and V4 terminals, and to the first motor controller **206A** through terminals V5 and V6.

(35) The mode switch **208** may switch the CTT system **100** between the acceleration mode and the deceleration mode, based on an instruction. To this end, the control module **108** of the CTT system **100** may include the receiver **210** that may be configured to wirelessly receive the instruction and transmit the instruction to the mode switch **208**. By way of an example, the receiver may be a wired or a wireless receiver, such as a Wi-Fi receiver, a Bluetooth receiver, etc. The receiver **210** may be

powered by the first motor controller **206A** (via terminals V7 and V8, for example, at 5 Volts). The receiver **210** may receive a wireless signal from an operator. Further, the receiver **210** may control the mode switch **208** (through terminals S1, S2) and control the second motor controller **206B** (through terminals S3, S4), and control the first motor controller **206A** (through S5, S6 terminals). (36) As such, the first motor controller **206A** may receive the electrical power from the voltage regulator **204** and a signal from the receiver **210** through terminals S5 and S6. Accordingly, the first motor controller **206A** may control the third stator **112A** in the acceleration mode or the first stator **106A** in the deceleration mode. At the same time, first motor controller **206A** may supply electricity (for example, 5 V), via terminals V7 and V8 to the receiver **210** to operate. The second motor controller **206B** may receive power from the voltage regulator **204** (via terminals V3 and V4) and a signal from the receiver ("via terminals S3 and S4), to thereby control the functioning of the second stator **110A**. It should be noted that the second stator **110A** may operate only in the acceleration mode, i.e. low torque application. In the first configuration (i.e. the acceleration mode), the first motor controller **206A** may be electrically coupled with the third stator **112A**, and the second motor controller **206B** may be electrically coupled with the second stator **110A**. As shown in FIG. 2A, in the first configuration (i.e. the acceleration mode), the mode switch **208** may electrically couple the rectifier **202** with the first stator **106A**. For example, the mode switch **208** may electrically couple the rectifier **202** with the first stator **106A**, by electrically coupling terminals I1, I2, I3 of the rectifier with terminals A1, A2, A3 of the first stator **106A**. Further, in the first configuration (i.e. the acceleration mode), the mode switch **208** may electrically couple the third stator **112A** with the first motor controller **206A**. For example, the mode switch **208** may electrically couple the third stator **112A** with the first motor controller **206A**, by electrically coupling terminals C1, C2, C3 of the third stator **112A** with terminals M1, M2, M3 of the first motor controller **206A**.

(37) As such, in the first configuration (i.e. the acceleration mode), the first stator **106A** may act as a generator and may supply generated alternating current (AC) to the rectifier **202** which may convert the alternating current (AC) into direct current (DC). Next, the direct current (DC) may enter the voltage regulator **204** which may bring the electrical supply to a constant voltage and send to the first motor controller **206A** and the second motor controller **206B**. Depending on the signal from the receiver **210**, electrical power may be sent to the second stator **110A** for low torque demand, or to the third stator **112A** for high torque demand.

(38) As mentioned above, the set of second planet gears **114C** may be attached to the connecting shaft **120** which may be coupled to a one-direction mechanism that allows rotation of the connecting shaft **120** and hence the set of second planet gears **114C** only in direction. In an embodiment, in the acceleration mode, the one way mechanism may allow the connecting shaft **120** to rotate in the same direction as the output shaft **118** and may restrict the rotation of the connecting shaft **120** in the opposite direction.

(39) It should be however noted that, in other embodiments, in the first configuration (i.e. the acceleration mode), the mode switch **208** may electrically couple the second stator **110A** (rather than the third stator **112A**) with the first motor controller **206A**.

(40) In the second configuration (i.e. the deceleration mode), the first stator **106A** may be coupled with the first motor controller **206A**. As shown in FIG. 2B, in the second configuration (i.e. the deceleration mode), the mode switch **208** may electrically couple the rectifier **202** with the third stator **112A**. For example, the mode switch **208** may electrically couple the rectifier **202** with the third stator **112A**, by electrically coupling terminals I1, I2, I3 of the rectifier with terminals C1, C2, C3 of the third stator **112A**. Further, in the second configuration (i.e. the deceleration mode), the mode switch **208** may electrically couple the first stator **106A** with the first motor controller **206A**. For example, the mode switch **208** may electrically couple the first stator **106A** with the first motor controller **206A**, by electrically coupling terminals A1, A2, A3 of the first stator **106A** with terminals M1, M2, M3 of the first motor controller **206A**.

(41) When the vehicle is decelerating, the energy of the vehicle may enter the CTT system **100** via the output shaft **118** from where it may be transferred to the engine, via the input shaft **102**. As such, in the deceleration mode, the third stator **112A** may work as a generator and the first stator **106A** as a motor. The current from the third stator **112A** may enter the rectifier **202**, then to the voltage regulator **204**, and then to the first motor controller **206A**. Thereafter, the first motor controller **206A** (with the help of the receiver **210**) may power the first stator **106A**, in a way to maintain a gear ratio that enables the input shaft **102** to run at a constant speed while the output shaft **118** slows down.

(42) Further, it should be noted that, in the deceleration mode, the one way mechanism may allow the connecting shaft **120** to rotate in the opposite direction as of the output shaft **118**, and may restrict rotation of the connecting shaft **120** in the other direction.

(43) Referring now to FIG. 3, a flowchart of a method **300** is illustrated, in accordance with some embodiments of the present disclosure. The method **300**, for example, may be performed by the control module **108**, as explained above.

(44) At step **302**, an instruction may be received to configure the CTT system **100** in one of an acceleration mode and a deceleration mode. The CTT system **100**, as explained in conjunction with FIG. 1, may include the first planetary gear assembly **104**, the first electric motor-generator **106**, the control module **108**, the second electric motor-generator **110**, the third electric motor-generator **112**, and the rectifier **202**. The CTT system **100** may be coupled with an input shaft **102** via the first planetary gear assembly **104**. The input shaft **102** may be attached to an engine of a vehicle implementing the CTT system **100**. The input shaft **102** may, therefore, supply mechanical power and rotate at a consistent speed. The first planetary gear assembly **104** may further include the first sun gear **104B** and the set of first planet gears **104C**. The first electric motor-generator **106** may include the first stator **106A** and the first rotor **106B**. The first electric motor-generator **106** may be implemented as an electromechanical device that can function both as a motor, converting electrical energy into mechanical energy, and as a generator, converting mechanical energy into electrical energy. The first rotor **106B** of the first electric motor-generator **106** may be attached to the first sun gear **104B** of the first planetary gear assembly **104**. Further, the first stator **106A** may be attached to the set of first planet gears **104C**. Furthermore, the first rotor **106B** may be electromagnetically coupled with the first stator **106A**. The control module **108** may be electrically connected to the first stator **106A** and the set of first planet gears **104C**. Further, the control module **108** may be free to rotate.

(45) The second electric motor-generator **110** may include the second stator **110A** and the second rotor **110B**. The second electric motor-generator **110** may also be implemented as an electromechanical device that can function both as a motor and as a generator. In an embodiment, the second rotor **110B** may be fixed to a housing associated with the CTT system **100**. As such, the rotation of the second rotor **110B** may be fixed. The second stator **110A** may be connected to each of the set of first planet gears **104C**, the first stator **106A**, and the control module **108**. The second rotor **110B** may be electromagnetically coupled with the second stator **110A**. The third electric motor-generator **112** may include the third stator **112A** and the third rotor **112B**. The third electric motor-generator **112** may also be implemented as an electromechanical device that can function both as a motor and as a generator. In some embodiments, each of the first electric motor-generator **106**, the second electric motor-generator **110**, and the third electric motor-generator **112** may be a brushless motor, and therefore, may require three electrical wires to power them. The third stator **112A** may be connected to the set of first planet gears **104C**, the first stator **106A**, the control module **108**, and the second stator **110A**. The set of first planet gears **104C**, the first stator **106A**, the control module **108**, the second stator **110A**, and the third stator **112A** may be coupled to an output shaft **118**. Furthermore, the third rotor **112B** may be electromagnetically coupled with the third stator **112A**.

(46) The rectifier **202**, as shown in FIGS. 2A-2B, may be electrically coupled to the control module

108, and may be configured to receive alternating current (AC) from an AC supply and convert it into direct current (DC). The control module **108** may configure the CTT system **100** in an acceleration mode and or a deceleration mode. In the acceleration mode, the first stator **106A** may act as a generator to generate electric power. The generated electric power may be supplied to the rectifier **202**. When the output shaft is accelerating (for example, when the vehicle starts to move from being stationary), the first stator **106A** (i.e. the first electric motor-generator **106**) may act as generator to generate electric power. As such, the first rotor **106B** is rotated by the mechanical power from the input shaft **102**, to thereby configure the first electric motor-generator **106** as generator to generate electrical power. The rectifier **202** may convert the alternating current (AC) into direct current (DC) that may be supplied to a voltage regulator **204**. The voltage regulator **204** may bring the current to a constant voltage and send the current to the first motor controller **206A** and the second motor controller **206B**. Depending on a signal from the receiver **210**, power may be sent to the second stator **110A** or the third stator **112A**. In particular, the power may be sent to the second stator **110A** for low torque demand, and to the third stator **112A** for high torque demand.

(47) During the deceleration mode, energy may be supplied to the CTT system **100** of the vehicle via the output shaft **118** and then to the engine via the input shaft **102**. The mode switch **208** may switch terminals. In the deceleration mode, the third stator **112A** may act as a generator to supply electric power to the rectifier **202**. The rectifier **202** may in turn supply the generated electric power to the first stator **106A**. In other words, when the output shaft is decelerating (for example, during braking of the vehicle), the third stator **112A** (i.e. the third electric motor-generator **112**) may act as generator to generate electric power. As such, the third rotor **112B** is rotated by the mechanical power from the output shaft **118**, to thereby configure the third electric motor-generator **112** as generator to generate electrical power. Further, in the deceleration mode, the first stator **106A** may act as a motor. Therefore, the first stator **106A** is supplied electrical power to cause rotation of the first rotor **106B**. As a result, a constant speed of the input shaft **104** is maintained during the deceleration of the output shaft **118**. The current generated from the third stator may enter the rectifier **202**, then to the voltage regulator **204**, and finally to the first motor controller **206A**. The first motor controller **206A**, with the help of the receiver **202** may power the first stator **106A** so as to maintain proper gear ratio, so that the input shaft **102** may have a constant speed while the output shaft **118** slows down.

(48) At step **304**, the mode switch **208** may be triggered to configure the CTT system **100** in one of an acceleration mode and a deceleration mode, based on the instruction. In the acceleration mode, the first stator **106A** may act as a generator to generate electric power that is supplied to the rectifier **202**. In the deceleration mode, the third stator **112A** may act as a generator to supply electric power to the rectifier **202**. The rectifier **202** may supply electric power to the first stator **106A** that acts as a motor to maintain a constant speed of the input shaft **102** during the deceleration of the output shaft **118**.

(49) Although the above detailed descriptions relate to specific preferred embodiments as the inventor presently contemplates, it will be understood that the invention in its broad aspects includes mechanical, chemical, and functional equivalents of the elements described herein. Various details of design and construction may be modified without departing from the true spirit and scope of the invention which is set forth in the following claims. Other embodiments, which will be apparent to those skilled in the art and which practice the teachings herein set forth, are intended to be within the scope and spirit of the invention.

Claims

1. A Continuous Torque Transmission (CTT) system comprising: a first planetary gear assembly comprising: a first ring gear attached to an input shaft rotating at a consistent speed; a first sun gear attached to a first rotor; and a set of first planet gears, each engaging with the first ring gear and the

first sun gear, a first electric motor-generator comprising: a first stator attached to the set of first planet gears; and the first rotor electromagnetically coupled with the first stator; a control module electrically connected to the first stator and the set of first planet gears; a second electric motor-generator comprising: a second stator connected to the set of first planet gears, the first stator, and the control module; and a second rotor electromagnetically coupled with the second stator; a third electric motor-generator comprising: a third stator connected to the set of first planet gears, the first stator, the control module, and the second stator, wherein the set of first planet gears, the first stator, the control module, the second stator, and the third stator are coupled to an output shaft; and a third rotor configured to electromagnetically coupled with the third stator; a rectifier electrically coupled to the control module and configured to receive alternating current (AC) and convert into direct current (DC); wherein the control module is to configure the CTT system in one of an acceleration mode and a deceleration mode, wherein, in the acceleration mode, the first stator acts as a generator to generate electric power that is supplied to the rectifier, and wherein, in the deceleration mode, the third stator acts as a generator to supply electric power to the rectifier, and the rectifier supplies electric power to the first stator that acts as a motor to maintain constant speed of the input shaft during the deceleration of the output shaft.

2. The CTT system of claim 1, wherein the second rotor is fixed to a housing associated with the CTT system to fix rotation of the second rotor.

3. The CTT system of claim 1 further comprising: a second planetary gear assembly comprising: a second ring gear attached to the set of first planet gears, the first stator, the control module, the second stator, and the third stator, and coupled to the output shaft; a second sun gear attached to the third rotor; and a set of second planet gears, each engaging with the second ring gear and the second sun gear.

4. The CTT system of claim 3, wherein the set of second planet gears is attached to a connecting shaft, wherein the connecting shaft is configured to rotate in one direction.

5. The CTT system of claim 3 further comprising: a first spur gear attached to the second ring gear, wherein the first spur gear is further coupled to the output shaft.

6. The CTT system of claim 5, wherein a second spur gear and a third spur gear are attached to a common connecting shaft (E) and a fourth spur gear (D) is attached to the output shaft, the fourth spur gear offset from the second spur gear and the third spur gear and engaging with the third spur gear.

7. The CTT system of claim 1, wherein, when the output shaft is stationary: the rotation of the input shaft causes the rotation of the first sun gear via the first ring gear and the set of first planet gears, to induce maximum voltage in the first stator, revolution of the set of first planet gears is constrained, and the control module ceases the supply of electrical power to the second stator or the third stator.

8. The CTT system of claim 1, wherein, when the output shaft is in the acceleration mode from being stationary: for a high torque application, the control module directs electrical power to the third stator, or for a low torque application, the control module directs electrical power to the second stator.

9. The CTT system of claim 1, wherein, when the output shaft rotates at a medium speed range: the torque generated at the output shaft is partly from the input shaft and partly from one of the second stator and the third stator.

10. The CTT system of claim 1, wherein, when the output shaft rotates at a high-speed range: the first sun gear rotates at a lower speed, the first rotor rotating at the lower speed induces a low voltage in the first stator, and the control module causes the supply of electrical power to one of the second stator and the third stator.

11. The CTT system of claim 1, wherein, when the output shaft rotates at a high-speed range and is in the deceleration mode, the output shaft rotates the third stator and the second stator, and the control module causes the supply of electrical power to the first stator.

12. The CTT system of claim 1, wherein the control module comprises: a mode switch configured to switch the CTT system between the acceleration mode and the deceleration mode, based on an instruction.
13. The CTT system of claim 12, wherein the control module further comprises a receiver configured to wirelessly receive the instruction and transmit the instruction to the mode switch.
14. The CTT system of claim 1 further comprising a voltage regulator coupled to the control module, the voltage regulator configured to receive electrical supply from the rectifier and regulate voltage of electricity supplied to the control module.
15. The CTT system of claim 1, wherein the control module comprises: a first motor controller configured to receive electrical power from a voltage regulator and the instruction from a receiver, to control the third stator in the acceleration mode, or control the first stator in the deceleration mode; and a second motor controller configured to receive electrical power from the voltage regulator and the instruction from the receiver, to control the second stator.
16. The CTT system of claim 15, wherein, in the acceleration mode, the first motor controller is coupled with the third stator and the second motor controller is coupled with the second stator wherein, in the deceleration mode, the first stator is coupled with the first motor controller and the second stator is coupled with the second motor controller.
17. The CTT system of claim 4, wherein, in the acceleration mode, the connecting shaft rotates only in the same direction as the output shaft, and wherein, in the deceleration mode, the connecting shaft rotates only in the opposite direction as of the output shaft.
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